Smart Web

Personal layouts for websites you never set up,
with Meta Muse Spark 1.3 and SAM 3.1

Technical report

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Abstract

Smart Web is a Chrome extension that reads a web page the user has never configured, asks Meta's Muse Spark 1.3 for a bounded layout plan, previews the plan on the real page and applies it reversibly. With the user's consent it learns which items they open and suggests emphasising them; the page changes only when they accept. The user can state what they come to the page for ("I only read the feed: hide both sidebars"); the plan hides, compacts, lays out as a grid, or emphasises regions of the site's own markup, never rewriting it. A screenshot can optionally be segmented with SAM 3.1 to give the planner visual groups. We built an evaluation set of 54 scenarios on 20 popular live websites with 124 checkable expectations, including 52 "keep" checks that catch over-hiding, and a runner that drives the extension through each scenario and checks the outcome on the live page. Three evaluation rounds found and fixed two real defects; in the last round Smart Web met 93% of expectations and fully completed 84% of scenarios, with no over-hiding, against 42% and 0% for leaving the page unchanged. Sending the screenshot to SAM made no consistent difference on this set. A round costs about $2.25 of Spark tokens at listed prices.

1 Problem

People use a handful of crowded websites every day — a code host, a Q&A site, a video site, a shop — and each serves the same distractions on every visit: marketing navigation, sign-up bars, recommendation columns, action buttons they never press. Fixing this today means writing CSS selectors in a user-style extension, which few people can do and which breaks when the site changes its markup. Reader modes help only on articles.

Smart Web lets a person say, in their own words, what they come to a page for, and turns that into a reviewed, reversible change to the page itself. The request that shaped the project came from the author's own use of GitHub, verbatim: *"please hide completly. i do not use at all the actions [unpin,watch,fork,start] neither on the top bar neitiher in the sidebar"*. It names the same controls in two places, which a single selector would miss.

Contributions.

- An MV3 extension that scans a page, obtains a validated Meta layout plan, previews it, applies it, restores the page, follows the tab across navigation, re-applies the layout when a single-page app redraws a region, and reuses saved plans on later visits and on other pages of the same kind without a model call (§2).
- Learning from use: opt-in, on-device counts of what the user opens, turned into suggestions that change the page only when accepted (§2).
- A planner contract: strict JSON schema, a local validator with one repair round, user goals, and follow-up questions whose answers adjust the plan locally (§3).
- An evaluation set of real sites and requests with machine-checkable expectations, a runner that measures outcomes on the live page, and three scored rounds (§4, §5).

2 System

Scan. On a toolbar click the extension injects a content script (Chrome's `activeTab`; no standing access to sites) that records the visible elements: tag, role, a coarse kind (landmark, interactive, list, item, heading,

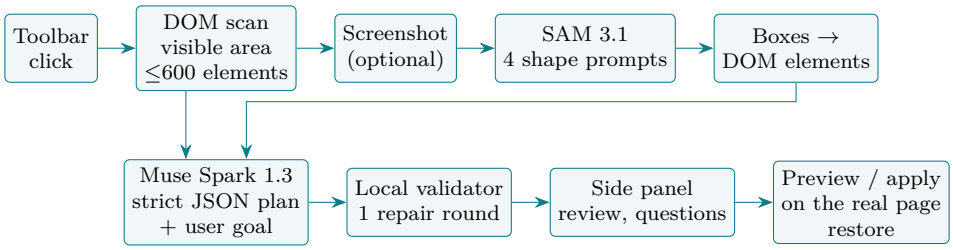


Figure 1: Pipeline. Meta calls go through a local LiteLLM proxy that holds the API key and a spend cap; the extension never sees the provider key.

media, group), parent, bounding box, up to 80 characters of visible text, link path and a few identifying attributes. Visibility uses `checkVisibility`, so closed menus and faded elements are left out. Form fields are listed without their contents and can never be targeted. Each element gets an ID valid only for this scan; the page refuses IDs from an older scan, so a plan can never mark the wrong element after the site re-renders.

Plan. Spark receives one line per element (`id <tag> kind parent=id [x,y,w,h] attrs href "text"`), the URL, the user's goal if given, and optionally SAM's visual groups. It returns, under a strict JSON schema, groups (a container, its members, a role and a reason), one operation per group from `hide-secondary`, `compact`, `grid`, `emphasize`, `none`, a URL pattern per group (site chrome gets a broad pattern, view-specific content a narrow one), and one to three follow-up questions whose options name the plan changes they imply. Answers are applied locally, without another call.

Validate and repair. A local validator rejects unknown or stale IDs, form fields, members outside their container, members claimed by two groups, grids of fewer than two items, oversized plans, URL patterns that do not match the page, and reorderable groups without a stable identity. If anything is rejected, Spark gets one repair round with the validator's messages; whatever still fails is dropped.

Apply and restore. The executor never moves or rewrites the site's DOM. It sets data attributes on the site's own elements and inserts one stylesheet keyed on them; restore removes both. Preview shows hidden regions dimmed and outlined in place, and Escape on the page undoes it. Links, focus order and the site's scripts keep working. A hide takes the plan's whole region only when the named members are most of it; otherwise only the members are hidden (§5).

Remember and follow. An applied plan is saved per page with node signatures; on the next visit, or on another page matching a group's URL pattern (e.g. `github.com/{owner}/{repo}/*`), it is remapped to the new scan and re-applied with no model call. Lists are re-found by row shape only when the plan covered the whole list. With the user's permission for a site, the extension follows the tab across navigation. "Hide something else" lets the user point at any element, widen or narrow the selection, and hide it; such hides are saved with the plan and survive a new plan.

Redraws. Single-page apps such as YouTube and GitHub replace whole regions when they re-render, which drops the marks. While a layout is kept, the page watches for marked elements leaving the document; once the burst settles the extension reads the page again and the saved plan re-applies, with no model call. A page that redraws more than five times in 30 seconds gets a notice instead, so a site that reacts to the change itself cannot loop.

Learning from use. Off until the user turns it on for a site. While a layout is kept, real clicks (`isTrusted`; scripted clicks are ignored) on items of its groups are counted on the device with a decaying per-item count. When one item has at least three uses and holds most of its group's weight, the panel asks: *"You opened 'Summit 45' 3 times. Highlight it in Backpack cards?"* Accepting adds a learned `emphasize` group over the plan, saved with it so it returns on later visits; either answer is remembered and the suggestion is not repeated; Reset clears what was learned. Learned changes are limited to emphasis: visually reordering a site's own controls would separate visual order from focus and reading order.

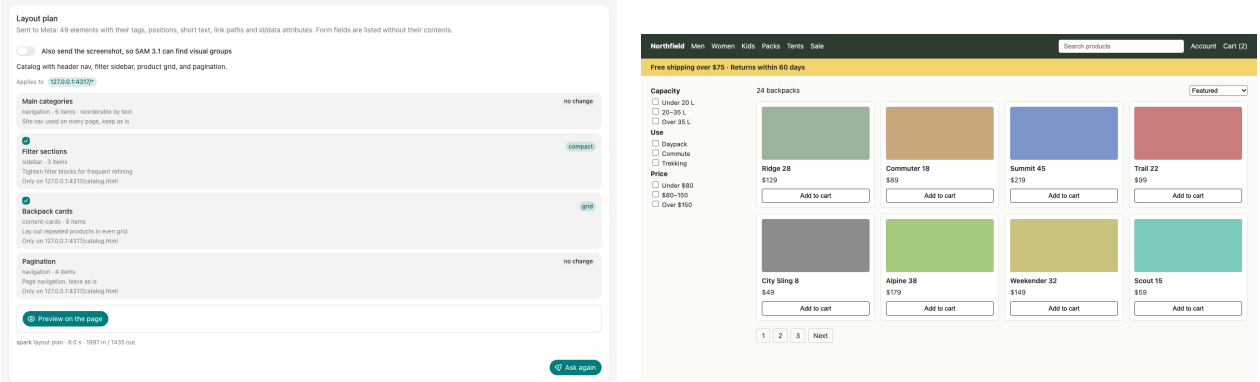


Figure 2: Owned test page. Left: the side panel with a live Spark plan (6.0 s): filters `compact`, product cards `grid`, navigation and pagination left alone, each with its reason and URL scope. Right: the same page after Keep; the cards were five and three per row before.

3 Use of Meta models

Muse Spark 1.3 does the planning: it reads an arbitrary page structure, names its regions, maps the user's goal onto them (including the same thing appearing in two places), chooses operations, writes URL patterns that generalise across pages of the same kind, and asks questions. Strict `json_schema` output held on every call, directly and through LiteLLM. Reasoning is about 70–90% of output tokens; `reasoning_effort: low` halved latency on our fixtures with the same plans and is the default. The first prompt produced "no change" for regions its own reasons called low-value; stating what a useful plan changes, and taking the user's goal, fixed this.

SAM 3.1 finds visually similar regions in the screenshot. Probing on UI screenshots showed it responds to shape phrases (*button*, *icon*, *rounded rectangle*, *gray box*) and returns nothing for UI names (*card*, *menu*, *panel*). The extension sends four shape prompts in parallel (about 1 s each), maps each box to the scanned element whose border box it best overlaps (IoU ≥ 0.5), and passes elements that look alike under one parent to Spark as visual groups. Section 5 measures whether this helps.

Proxy. A local LiteLLM proxy (`meta/muse-spark-1.3`, `meta/sam-3.1`) holds the API key and a $100 budget; the extension holds only a local key. SAM's `/v1/responses` passes through LiteLLM unchanged.

4 Evaluation

4.1 Evaluation set

Twenty pages from popular sites of different kinds (Table 1), all loadable in an automated browser logged out. Six sites first chosen (X, Reddit, Hacker News, npm, IMDb, eBay) refuse automated browsers and were replaced by sites of the same kind. Each scenario is what a user types into "I come here to…", or, for six picker scenarios, the element they click. Expectations say what must be hidden, emphasised, compacted or laid out as a grid, and what must *stay* visible. In total: 60 hide, 52 keep, 8 emphasise, 3 compact and 1 grid expectations. No scenario hides a sign-in wall or cookie prompt; one hides an ad slot.

Expectations name elements by *anchors* that survive markup changes: tag, kind, text or text prefix (case, whitespace and digits ignored, so counters do not break them), link path, exact attributes, or a viewport position range. A validator resolves all 130 anchors against collected scans and rejects loose anchors and scenarios that ask to keep something inside a region they hide. Every anchor was also checked by eye; where only a region's heading was anchorable, a second element of the region is required too.

4.2 Runner and scoring

The runner (Playwright, the extension's test build) opens each live site, answers cookie prompts with "Reject", pauses media, scans, and resolves the anchors against *this visit's* scan. It then sends the goal twice on the same scan — DOM only, then DOM plus screenshot — applies every proposed change with the plan's default answers, inspects the anchors on the live page through a read-only content-script probe, and restores. An

Site	Page	Goal scenarios	Picker	Expectations
YouTube	video feed	2	1	8
YouTube	video page	3	0	9
Twitch	stream directory	3	1	8
Twitch	category streams	1	0	3
GitHub	code repository	3	1	9
Stack Overflow	Q&A thread	3	1	9
Wikipedia	encyclopedia article	3	0	8
Amazon	product search	3	0	6
MDN	documentation	2	0	5
BBC News	news front page	2	0	3
LinkedIn	job search	1	0	3
The Verge	news front page	2	0	4
Spotify	music artist page	2	0	5
Booking.com	travel search	2	0	4
Bluesky	social profile	3	0	6
DEV Community	community feed	3	0	7
The Guardian	news front page	3	0	6
PyPI	package page	2	0	6
Steam	game store page	3	1	8
IKEA	product listing	2	1	7

Table 1: Evaluation set: 20 pages, 54 scenarios, 124 expectations. Collected logged out from Switzerland at 1280×800 on 28 October 2026.

Run	Mode	Expectations met	Scenarios fully passed	Hidden as asked	Kept (no over-hiding)	Emph./compact/ grid
Do nothing	–	42% (41/97)	0% (0/42)	0% (0/46)	100% (41/41)	0% (0/10)
Round 1	DOM	84% (81/97)	64% (27/42)	83% (38/46)	85% (35/41)	80% (8/10)
	DOM+SAM	78% (76/97)	60% (25/42)	78% (36/46)	80% (33/41)	70% (7/10)
Round 2	DOM	87% (92/106)	71% (32/45)	73% (37/51)	100% (45/45)	100% (10/10)
	DOM+SAM	89% (94/106)	76% (34/45)	78% (40/51)	100% (45/45)	90% (9/10)
Round 3	DOM	93% (94/101)	84% (37/44)	87% (41/47)	100% (44/44)	90% (9/10)
	DOM+SAM	93% (94/101)	84% (37/44)	87% (41/47)	100% (44/44)	90% (9/10)
Round 4	DOM	93% (100/108)	83% (38/46)	86% (44/51)	100% (46/46)	91% (10/11)

Table 2: Goal scenarios, Spark effort low. Rounds 1–3 on 28 October 2026; round 4 on 29 October, DOM only, a regression check after adding redraw handling and learning. The six picker scenarios passed every scored expectation in every round. Denominators vary because sites change between visits; unscored expectations are excluded.

element counts as shown if it is visible and still shows text or media; a wrapper whose contents were all hidden counts as hidden.

- **hide** passes when the element is no longer shown; **keep** when it still is.
- **emphasize/compact/grid** pass when the element, an ancestor, or a descendant covering at least half of it carries that operation.
- Not scored: anchors that match nothing on that visit (the site changed or served a bot check), elements not shown before any change, and scenarios where the site navigated mid-run.
- **Do nothing** is the untouched page, scored the same way.

5 Results

Table 2 gives the rounds. The extension changed between rounds, so they are a before/after sequence, not repeats.

Round 1 met 84% of expectations (DOM only) but hid things the user wanted kept in 6 of 41 keep checks. All six had one cause: the plan named a wide container for a narrow request (the whole left menu as the container for "Shorts") and the executor hid the container. "Remove Shorts from the left menu" removed Home and Subscriptions too; hiding Steam's sign-in banner took the game's title. The same widening existed in saved plans: a one-item group was saved as "every row of this shape" and would have come back as the whole list.

Run	Mode	Plans	Repaired	Spark calls	Input tokens	Output tokens	of which reasoning	Median plan (s)	SAM calls	Spark cost
Round 1	DOM	45	25	70	401k	139k	100k	12.3	0	$1.09
	DOM+SAM	45	21	66	368k	139k	103k	13.2	180	$1.05
Round 2	DOM	45	24	69	409k	136k	100k	13.0	0	$1.09
	DOM+SAM	45	27	72	423k	146k	108k	14.8	180	$1.15
Round 3	DOM	44	23	67	409k	139k	101k	13.2	0	$1.10
	DOM+SAM	44	25	69	423k	147k	109k	14.4	176	$1.15
Round 4	DOM	46	23	69	415k	139k	102k	12.3	0	$1.11

Table 3: Cost and speed per round and mode. Spark cost uses the listed prices of 27 October 2026 ($1.25 per million input, $4.25 per million output tokens, reasoning counted as output); SAM's price was not established. The proxy keeps no spend records, so these are computed from token counts, not billed amounts.

Round 2 hid only the named members unless they were all of the region. Over-hiding went to zero, but hiding fell to 73%: plans name a list's links but not its heading or toggle button, so the links went and an empty box stayed (Wikipedia's contents, Stack Overflow's and Amazon's navigation).

Round 3 hides the region when the members are most of it (more than half of its visible scanned children, counted recursively). It met 93% of expectations and completed 84% of scenarios in both modes, with no over-hiding. Part of the gain is in measurement, not the extension: the shown-means-has-content rule turned GitHub's marketing menu into a pass (it was gone from the screen in every round), and a second BBC consent prompt that had covered the page is now answered.

Remaining failures (7 per mode in round 3): two on Booking.com, whose results do not render in the automated browser; four partial hides where the plan's container sits one level inside the region, leaving its heading (Twitch's channel sidebar, Steam's tags, Bluesky's trending column and sign-up bar); and one emphasis or compaction that varied between plans. PyPI served a bot check and two scenarios' elements were absent that visit, so they were not scored.

Does SAM help? The screenshot mode scored −6, +2 and 0 points against DOM only across the three rounds, for 176–180 extra SAM calls per round. On this set we see no measured benefit. Pages here are large, well-structured documents whose DOM already carries the grouping; SAM may matter more on canvas-heavy or div-soup pages, which the set does not cover.

Cost and speed (Table 3). A plan takes a median 13–15 s and about 9 k input tokens; a full round of 88–90 plans costs about $2.25 in Spark tokens and 9–13 minutes with three browsers. About half the plans need a repair round, which is the largest remaining cost to cut. A saved plan costs nothing on later visits.

6 Limitations

- **One annotator, one locale, logged out.** Expectations are one reasonable reading of each request, written by the author from scans and screenshots, not agreed by several annotators. Logged-in pages differ; a signed-in GitHub top bar is a known gap.
- **Visible area only.** The scan covers the viewport, so every anchor is above the fold at 1280×800.
- **Small numbers for some operations.** Emphasis, compaction and grid have 12 expectations together; their rates rest on few cases.
- **Live sites drift.** Each visit is resolved against a fresh scan, but sites change content, run experiments and serve bot checks; rounds are not exact repeats.
- **Developed against its own test set.** The fixes between rounds were made after seeing failures on these scenarios. They address general mechanisms (how a hide is scoped), not individual sites, but a held-out set would give a fairer number.
- **Outcome, not benefit.** We measure whether the page changed as asked, not whether people complete tasks faster. No user study has been run.

7 What is new, and what was reused

All extension code, the planner contract, the evaluation set and runner were written during the event (from 28 October 2026). The extension started from the author's own MIT starter template (`tymzar/starter-web-extension`, imported unmodified as a separate commit). The repository also contains the author's earlier adaptive-interface work (a local ranker and CSS-computed inference, last changed 23 October 2026). Learning reuses one part of it, the decaying per-item count (`src/model/affinity.ts`), unchanged; the organisers confirmed on 29 October that reusing credited pre-event code is allowed. The neural ranker itself is not used: it learns from pointer signals the extension does not collect.

8 Next steps

Planned before submission: trying the extension on more page families and recording the demo. If time remains: hide the heading of a hidden region; reduce repair rounds by fixing the most common validator rejections at the prompt; and optimise the planner prompt against this evaluation set with DSPy, after making scoring cheap by evaluating plans offline against the collected scans and holding out sites.

Feedback for Meta (from our running log). SAM's docs are hard to find (the obvious URL returns 404), a zero-match concept returns an empty `output_text` that looks like an error, and a note that SAM answers shape phrases rather than UI names would help screen use cases. LiteLLM's Meta docs lag the model list and omit SAM, which works through it. A per-request latency hint, or a faster structured-output tier, would help interactive use, where reasoning dominates output.

A All scenarios with round-3 results

One mark per expectation in order: ✓ pass, ✗ fail, ? element not on the page that visit, – not shown before any change, n/s not scored (site navigated mid-run). Picker scenarios make no model call; one result covers both columns.

Site	What the user asks	Expectations	DOM	DOM+SAM
YouTube	I just want to watch the video. Hide the recommended videos column.	hide keep keep	✓✓✓	✓✓✓
YouTube	I never like, dislike or share videos. Hide those buttons but keep Subscribe.	hide hide hide keep	✓✓✓✓	✓✓✓✓
YouTube	Make the video title and channel stand out; recommendations can be denser.	emph compact	✓✓	✓✓
YouTube	I never watch Shorts. Remove Shorts from the left menu.	hide keep keep	✓✓✓	✓✓✓
YouTube	Hide the 'history is off' box; I only use search.	hide hide keep	✓✓✓	✓✓✓
Twitch	I only browse categories. Hide the Live Channels sidebar on the left.	hide keep keep	✗✓✓	✗✓✓
Twitch	Hide the sticky 'Join the Twitch community' bar at the bottom.	hide keep	✓✓	✓✓
Twitch	Fit more categories on screen.	compact keep	✓✓	✗✓
Twitch	Hide the 'May Contain Labeled Content' notice and the sign-up footer.	hide hide keep	✓✓✓	✓✓✓
GitHub	please hide completly. i do not use at all the actions [unpin,watch,fork,start] neither on the top bar neitiher in the sidebar	hide hide keep	✓✓✓	✓✓✓
GitHub	Hide GitHub's marketing menu at the top (Platform, Solutions, Resources…).	hide keep	✓✓	✓✓
GitHub	review pull requests and check Actions	emph keep	✓✓	✓✓
Stack Overflow	I read answers only. Hide the right sidebar (blog, meta, promos).	hide keep keep	✓✓✓	✓✓✓
Stack Overflow	Hide the left navigation; I arrive from search and never use it.	hide keep	n/s	n/s
Stack Overflow	Hide the 'Stack Overflow for Agents' promo but keep the rest of the sidebar.	hide keep	✓✓	✓✓
Wikipedia	Distraction-free reading: hide the Appearance panel and the Contents sidebar.	hide hide keep keep	✓✓✓✓	✓✓✓✓
Wikipedia	Hide Donate, Create account and Log in; keep the search box.	hide keep	✓✓	✓✓
Wikipedia	I jump between sections a lot; make the table of contents stand out.	emph keep	✓✓	✓✓
Amazon	Hide the left filter column; I just scan results.	hide hide keep	✓✓✓	✓✓✓
Amazon	Hide the second navigation bar (All, Today's Deals, …); keep the search bar.	hide keep	✓✓	✓✓
Amazon	Show the results as a compact grid so I can compare more at once.	grid	✓	✓

Site	What the user asks	Expectations	DOM	DOM+SAM
MDN	Hide the left reference sidebar but keep 'In this article'.	hide keep keep	✓✓✓	✓✓✓
MDN	Hide the Baseline compatibility box at the top.	hide keep	✓✓	✓✓
BBC News	Hide the second row of the navigation (US & Canada, UK, Africa. . .).	hide keep	✓✓	✓✓
BBC News	Make the top story block stand out.	emph	?	?
LinkedIn	Hide 'Get notified when a new job is posted' and the 'Use AI to assess how you fit' box.	hide hide keep	??✓	??✓
The Verge	Hide the Subscribe and Sign In links.	hide hide keep	✓✓✓	✓✓✓
The Verge	Make the top navigation more compact.	compact	✓	✓
Booking.com	I only book stays. Hide the Flights / Car rental / Cruises tabs.	hide keep	✗✓	✗✓
Booking.com	Hide the big footer link lists.	hide keep	✗✓	✗✓
Spotify	Hide Premium, Support, Download and Install App in the top bar.	hide hide keep	✓✓✓	✓✓✓
Spotify	Hide the Now playing panel on the right.	hide keep	✓✓	✓✓
GitHub	*Clicks the element:* , data-testid = repo-header-actions	hide keep	✓✓	
YouTube	*Clicks the element:* <a>, aria-label = Shorts	hide keep	✓✓	
Stack Overflow	*Clicks the element:* <aside>, aria-label = Stack Overflow for Agents	hide keep	✓✓	
Twitch	*Clicks the element:* id = twilight-sticky-footer-root	hide	✓	
Bluesky	Hide the Trending column on the right; I only read this profile's posts.	hide hide keep	✗✓✓	✗✓✓
Bluesky	Hide the Create account / Sign in bar at the bottom.	hide keep	✗✓	✗✓
Bluesky	I switch between Posts, Media and Videos a lot; make those tabs stand out.	emph	✓	✓
DEV Community	Hide the right sidebar with livestreams and listings.	hide keep	✓✓	✓✓
DEV Community	I only read the feed: hide both sidebars.	hide hide keep	✓✓✓	✓✓✓
DEV Community	Hide Log in and Create account in the top bar; keep the search.	hide keep	✓✓	✓✓
The Guardian	Hide the ad space at the very top and the Print subscriptions link.	hide hide keep	–✓✓	–✓✓
The Guardian	Hide the 'You are viewing the International homepage' banner.	hide keep	✓✓	✓✓
The Guardian	Make the highlighted stories row stand out.	emph	✓	✓
PyPI	Hide the announcement bar at the top.	hide keep	??	??
PyPI	I come here for the install command; make it stand out and hide the project sidebar.	emph hide hide keep	????	????
Steam	Hide the 'You're not signed in! Open in Desktop App' banner.	hide keep	✓✓	✓✓
Steam	Before buying I check reviews; make the review summary stand out.	emph	✗	✓
Steam	Hide the Install Steam button and the user-defined tags.	hide hide keep	✓✗✓	✓✗✓
IKEA	Hide the 15% off / IKEA Family promotion block.	hide keep	✓✓	✓✓
IKEA	Hide the chat button and the postcode / store pickers.	hide hide keep	✓✓✓	✓✓✓
Steam	*Clicks the element:* , text starting "Open in Desktop App You're not signed in"	hide keep	✓✓	
IKEA	*Clicks the element:* aria-label = Open chat window	hide keep	✓✓	